$$\sum_{n=n_0}^{\infty} a_n$$

① Confronto asintotico

② Leibniz $a_n \searrow 0 \Rightarrow \sum (-1)^n a_n$ CONVERGE

③ rapporto $\frac{a_{n+1}}{a_n} \to l$ $l<1$ CONVERGE

$l>1$ DIVERGE

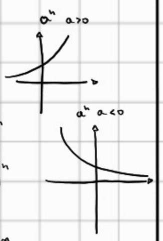
④ radice $\sqrt[n]{a_n} \to l$ $l=1$?

① $\sum_{n=1}^{\infty} \frac{\cos n^4}{n^2+1}$ CONV.

$\left[\int_{100}^{\infty} \frac{\cos x^4}{x^2+1}\right]$ ↑ qualsiasi valore

$$\left|\frac{\cos n^4}{n^2+1}\right| \le \frac{1}{n^2+1} \le \frac{1}{n^2} \xrightarrow[\text{di ordine 2}]{n\to\infty} 0$$

$\cdot \sum_{n=1}^{\infty} 2^n \cdot \sin\left(\frac{1}{3^n}\right)$



$$2^n \cdot \sin\frac{1}{3^n} \overset{n\to\infty}{\sim} 2^n \frac{1}{3^n} = \left(\frac{2}{3}\right)^n = \left(\frac{3}{2}\right)^{-n}$$

$(\sin x \sim x \quad x \to 0)$

$\xrightarrow{n\to\infty} 0$ esponenzialmente

$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ $a_n = \left(\frac{2}{3}\right)^n$

$$\sqrt[n]{a_n} = \sqrt[n]{\left(\frac{2}{3}\right)^n} = \boxed{\frac{2}{3} < 1}$$

$n! \sim \sqrt{2\pi n} \cdot \left(\frac{n}{e}\right)^n$

$\cdot \sum_{n=1}^{\infty} \underbrace{\frac{(2n)!}{n^{2n}}}_{a_n}$ $\boxed{\text{CONVERGE}}$ $n! = n(n-1)!$

$$\frac{a_{n+1}}{a_n} = \frac{[2(n+1)!]}{(n+1)^{2(n+1)}} \cdot \frac{n^{2n}}{2n!} = \frac{(2n+2)(2n+1)\cancel{(2n)!}}{(n+1)^{2n}(n+1)^2} \cdot \frac{n^{2n}}{\cancel{2n!}} =$$

$$= \frac{2(n+1)(2n+1)}{(n+1)^2} \cdot \left(\frac{n}{n+1}\right)^{2n} \to 4e^{-2}$$

$\frac{2(n+1)(2n+1)}{(n+1)^2} \to 4$

$\left(\frac{n}{n+1}\right)^{2n} = \left(\frac{n+1}{n}\right)^{-2n} = \left(\underbrace{\left(1+\frac{1}{n}\right)^n}_{\to e}\right)^{-2}$

$\frac{4n^2+6n+2}{n^2+2n+1}$

$\sim \frac{4n^2}{n^2} = 4$

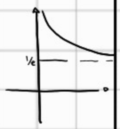
$\cdot \sum_{n=1}^{\infty} \overbrace{\left(1-\frac{1}{n^2}\right)^{n^2}}^{a_n}$

$\ln(1+x) = x + o(x^2)$ $x \to 0$

$$a_n = \exp\left(n^2 \cdot \ln\left(1-\frac{1}{n^2}\right)\right) = \exp\left(n^2 \cdot \left(-\frac{1}{n^2} + o\left(\frac{1}{n^2}\right)\right)\right) =$$

↑ $e^{(\dots)} = \exp(\dots)$

$$= \exp\left(-1 + o\left(\frac{1}{n^2}\right)\right) = e^{-1+o\left(\frac{1}{n}\right)} \to e^{-1} = \frac{1}{e}$$



Per le serie armoniche → $\sum_{n=1}^{\infty} \frac{1}{n}$ DIVERGE

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ CONVERGE

② $\sum_{n=1}^{\infty} (-1)^n \frac{2^n+n}{3^n+n^2}$ → $|a_n| = \frac{2^n+n}{3^n+n^2} \sim \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n \to 0$ esponenziale CONV ASS.

↓ MA

$\sum_{n=1}^{\infty} (-1)^n \frac{2+n}{1+n+n^2}$

$|a_n| = \frac{n+2}{n^2+n+1} \sim \frac{1}{n} \to 0$ ord $\boxed{1}$ DIVERGENTE

o almeno, non converge ASSOLUTAMENTE

$\frac{2+n}{n^2+n+1} \to 0$ ok

non serve decrescente per tutto, ma che sia decrescente in un punto "alto"

$\frac{2+n}{n^2+n+1} \sim \frac{n}{n^2} = \frac{1}{n} \searrow$ ↑ decrescente

Per il criterio di Leibniz non converge assolutamente, ma converge semplicemente

$\cdot \sum_{n=1}^{\infty} \left(\frac{\ln(n)}{n}\right)^2$ CONV. $\frac{(\ln(n))^2}{n^2} \to 0$ di ordine $2-\varepsilon$ $\forall \varepsilon$

$$\frac{(\ln(n))^2}{n^2} \le \frac{\sqrt{n}}{n^2} = \frac{1}{n^{3/2}} \to 0 \quad \text{ord } 3/2$$

$n^{1/2}$ è arbitrario, qualsiasi cosa, basta che non sia ①

$(\ln(n))^2 \le n^{\varepsilon}$ $\forall \varepsilon$

$\frac{(\ln(n))^2}{n^2} \le \frac{n^{\varepsilon}}{n^2} \to 0$ ord $2-\varepsilon$

$\cdot \sum_{n=2}^{\infty} \left(\frac{n+1}{n-1}\right)^n$

$\sqrt[n]{a_n} = \frac{n+1}{n-1} \to 1$?

$$\left(\frac{n+1}{n-1}\right)^n = \exp\left(n \cdot \ln\left(\frac{n+1}{n-1}\right)\right) = \exp\left(n \cdot \ln\left(1+\frac{2}{n-1}\right)\right) \sim e^2 \ne 0$$

$\frac{n+1}{n-1} = \frac{n-1+2}{n-1}$

$\frac{2}{n-1} \sim \frac{2}{n}$

$\cdot \sum_{n=1}^{\infty} \frac{\ln(n!)}{n^3}$ CONV.

log rallenta le cose e $n!$ va velocissimo

→ $\ln(e^n) = n$

$\ln(n) \le n^{0{,}001}$

buttiamo via un po' di informazioni

$$n! = n \cdot (n-1)(n-2)(n-3) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$$

↑ n termini

$$\le n \cdot n \cdot n \cdot n \ldots \cdot n \cdot n \cdot n \le n^n$$

$$\frac{\ln(n!)}{n^3} \le \frac{\ln(n^n)}{n^3} = \frac{n\ln(n)}{n^3} = \frac{\ln(n)}{n^2} \to 0 \quad \text{ord} > 3/2$$

$\cdot \sum_{n=1}^{\infty} \frac{1}{n\sqrt[n]{n!}}$ CONV? $\frac{1}{n\sqrt[n]{n!}} \le \frac{1}{n \cdot n^{1/2}} \to 0$ ord $3/2$

$$n! = n(n-1)(n-2)(n-3) \cdot \ldots \cdot 4 \cdot 3 \cdot 2 \cdot 1 =$$

$$= n(2(n-1))(3(n-2))(4(n-3)) \cdot \ldots$$

↑ $\frac{n}{2}$ termini

$$\ge n \cdot n \cdot n \cdot \ldots \cdot n = n^{n/2}$$

↑ non è ottimale questa stima

$$\sqrt[n]{n!} = \sqrt[n]{n^{n/2}} = n^{1/2}$$

$$\sum_{n=1}^{N} n = \frac{N(N+1)}{2}$$ ← Gauss

Il Polinomio di Taylor,

③ $\sum_{n=1}^{\infty} \frac{x^n}{n 2^n}$ Studiare il carattere della serie al variare di $x \in \mathbb{R}$

$\frac{x^n}{n2^n} = \frac{1}{n}\left(\frac{x}{2}\right)^n$ → qualcosa alla n può andare da 0 a ∞

↑ decadimento di ordine 1

$a^n \to \begin{cases} 0 & a<1 \\ \infty & a>1 \end{cases}$ prendiamo $a>0$

. Se $\frac{x}{2} > 1$ cioè $x > 2$

allora $\frac{1}{n}\left(\frac{x}{2}\right)^n \to \infty$

DIVERGENDO A $+\infty$ $\sum \frac{x^n}{n2^n}$

Per lo stesso ragionamento

. Se $\left|\frac{x}{2}\right| < 1$ cioè $|x| < 2$

allora $\frac{1}{n}\left(\frac{x}{2}\right)^n \to 0$ esponenzialmente

quindi $\sum \frac{x^n}{n2^n}$ CONV. ASS.

. Se $x < -2$

la serie è indet.

$\sum (-1)^n \underbrace{a_n}_{\to +\infty}$

. Se $x = 2$

$$\sum_{n=1}^{\infty} \frac{x^n}{n2^n} = \sum_{n=1}^{\infty} \frac{1}{n} \text{ DIV (serie armonica)}$$

. Se $x = -2$

$$\sum_{n=1}^{\infty} \frac{(-2)^n}{n2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$ CONV. SEMPL. MA NON ASS.

$\sum_{n=1}^{\infty} \frac{(n+1)(2+\sin(n))}{\sqrt[3]{n^5}}$ div.

$\boxed{\sin n \sim n \quad n \to 0}$

$\frac{(n+1)(2+\sin(n))}{n^{5/3}}$ → limitato

$\sum_{n=1}^{\infty}$

④ $\sum_{n=0}^{\infty} (-1)^{n+1} \frac{(\ln(10))^n}{n!}$

$\left|(-1)^{n+1}\frac{(\ln(10))^n}{n!}\right| \ge \frac{(\ln(10))^n}{n^{1/2}}$

CONV.

$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$

$|x| < 1$

HAPPY BIRTHADAY TO THE GROUND!